MALNAD COLLEGE OF ENGINEERING, HASSAN

(An Autonomous Institution affiliated to VTU, Belagavi)

24CS503(Computer Networks) Mini Project Synopsis On

"ARP MONITOR & ANOMALY DETECTOR"

Submitted in partial fulfillment of the requirements for the award of the degree of

“Bachelor of Engineering in Computer Science and Engineering”

Submitted by:

1.H B Sankalp 4MC24CS069

2.Kusuma K R 4MC25CS411

Under the guidance of:

Dr.Ananda Babu J

Prof and HOD

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

2026-2027

ARP MONITOR AND ANOMALY DETECTOR

1. Abstract

Address resolution protocol ARP maps an IP address to its corresponding MAC address on a local network. Since these mappings may change over time, it would help if we monitored them for any changes or abnormalities. The ARP Monitor and Anomaly Detector captures ARP packets using Scapy and Npcap to extract relevant details such as IP address and MAC address of each packet then stores these records in an SQLite database file. Any change of an existing IP with a different MAC address would be logged here as possibly abnormal information that needs more scrutiny from us. Charts are used for flash dashboard to display statistics such as traffic information and events detected by them. Normal and abnormal test results show what happens when you run an application on it.

2.Introduction

The Address Resolution Protocol (ARP) is used for finding the physical device identifier of an IP address on a LAN. It enables them for delivering frame at proper location within a LAN.

Since an IP MAC link may be changed over time, it helps to track ARP traffic so that any unusual behavior could be detected quickly. Instead of checking the ARP table manually at each check, this proposal maintains past records to compare them later on.

ARP Monitor and Anomaly Detector is an application that monitors for ARPs and detects anomalies by extracting IP addresses from them to store their corresponding MAC address details along with any other available data on those devices' network traffic patterns over time. The web dashboard shows out information about it.

Scapy and Npcap are used to capture packets while SQLite is utilized as a database of information whereas Flask along with Chart.

There are two experimental studies to be conducted here. Normal cases keep a consistent IP to MAC mapping whereas abnormal ones change it of any given IP. It is an easy method for demonstrating detection algorithms within a controlled setting.

3. Problem Statement

ARP monitor software is available for viewing ARP request, reply information as well as an up-to-date list of all active tables on a computer system. Nonetheless, it might be difficult to

maintain an accurate history of past IP-MAC associations for reference purposes. If there is an abundance of ARP event generation then manual inspection might be hard to do. For simplicity purposes, our model collects ARP packet data from network devices to track their mapping over time; it also detects any deviations between current observations and previous ones before storing them on disk so that they can be viewed via an integrated interface.

4. Existing System

Address Resolution Protocol (ARP) data is available to monitor from an application through capture software for packets on your computer; you may also find it within OS's ARP tables of devices connected to that machine. ARP request/response monitoring software allows for observing these events while also showing current IP-to-MAC address mapping information of that machine. Packet analyzer provides details on each packet of ARP like source address and destination address whereas OS utility shows live ARP table.

While these techniques can be helpful to diagnose simple problems of a computer network through monitoring packets but do not provide continuous analysis thereof. Generally speaking, you need to check captured frames or arp table records individually for determining if there is any correlation between IP addresses and MAC addresses. It is hard for us to keep track of these things on an ongoing basis as well as identifying them at different points in time.

Also traditional methods don't have an explicit way to store each detected ARP packet and compare them against previous records of mapping data. In case of changes, If there is any change then an admin has to do manual comparison between present data and past records. It may take some time to do so because there are many networks that have several computers generating lots of ARP messages.

A further issue would be that there are no interfaces to combine packet monitoring, history storage, anomaly detection statistics analysis and visualization tools together. There might be different types of software needed to capture packets, store databases, analyze them and graphically represent them. Therefore, data could be spread out among several utilities.

Limitations of Existing Systems:

- Manual, ARP data might require manual inspection.
- IP-MAC historical data may be stored separately from other databases.
- SusBY IP MAC suspicious changes might need to be compared manually.
- Anomaly detection is not available through basic monitoring system.
- Analysis of statistics and graphs might need different software programs.

- No such thing exists for a particular application to monitor it all at once like this. make it little short
- No centralized monitoring dashboard.

5. Proposed / New System

ARP Monitor and Anomaly Detector is a version of standard ARP monitoring which has real time packet capturing capability, history of ip-mac mapping database auto change detector software storage information statistics visualization capabilities over internet accessible through web interface. The main purpose is to provide live tracking of ARP activity statistics along with identifying irregularitiesity among device MAC addresses within an organization's computer networks without requiring them from doing it manually by themselves.

The ARP request/response is collected from that machine using a specific interface of an NPCAP / Scapy program for capturing it. ARP packets are captured by Scapy on a machine that uses npcap to python library for networking tasks. Each packet's information such as its IP address, MAC address, ARP operation and time of capture would be analyzed by us.

Data is collected from detectors to monitor it later on. This system tracks current IP addresses and their corresponding MAC address for comparison purposes purpose. When a MAC address is found on more than one IP address that, it would be classified as an anomaly according to this type of system. This allows for automatic identification of differences between an IP address and a MAC address through it.

The information of this system will be stored in a database SQLite file for monitoring ARP addresses and identifying anomalies. Data anomalies are saved at this place.

IP address

Previous MAC address

New MAC address

Timestamp

Severity

Description of the detected change

The historical records of this project allow for comparison between currents present day information on a map from earlier times to help study network change over time. It also does not depend on any pre-existing ARPAN database for its operation.

Flask-based web app is available on this or off of your computer via a browser. The dashboard shows collected ARP information informats in tabular form as well as bar charts to make it easier to view them. ARP traffic statistics such as total 1sar data set; unique IP address count; unique MAC address counts etc. fall under anomaly detection categories too. Here is a brief introduction of this network for you.

The system also uses Chart.js to visualize ARP activity and anomaly-related statistics. Graphical representation allows for, a better understanding about change in network traffic pattern and also to identify correlations between regular ARP events' data point and anomaly detection result. The dashboard is, being an automated system that automatically updates its information so it can display recent ARP events without requiring manual refreshment by humans.

For demonstration purposes purpose so that we can test this out by simulating normal and abnormal cases. Usually there are no differences between IP addresses and MAC addresses while they change from each other under normal circumstances of a particular IP address. Simulation is used to test detectors through simulations rather than real-world tests.

Therefore, thus it includes each stage from capturing ARP packets to extracting them as packets then MAC addresses followed by historical analysis and anomalies detection; storing databases and retrieving statistics via web browsers. An efficient method for monitoring ARP traffic and detecting any abnormal IP-MAC address associations within a local network environment of yours.

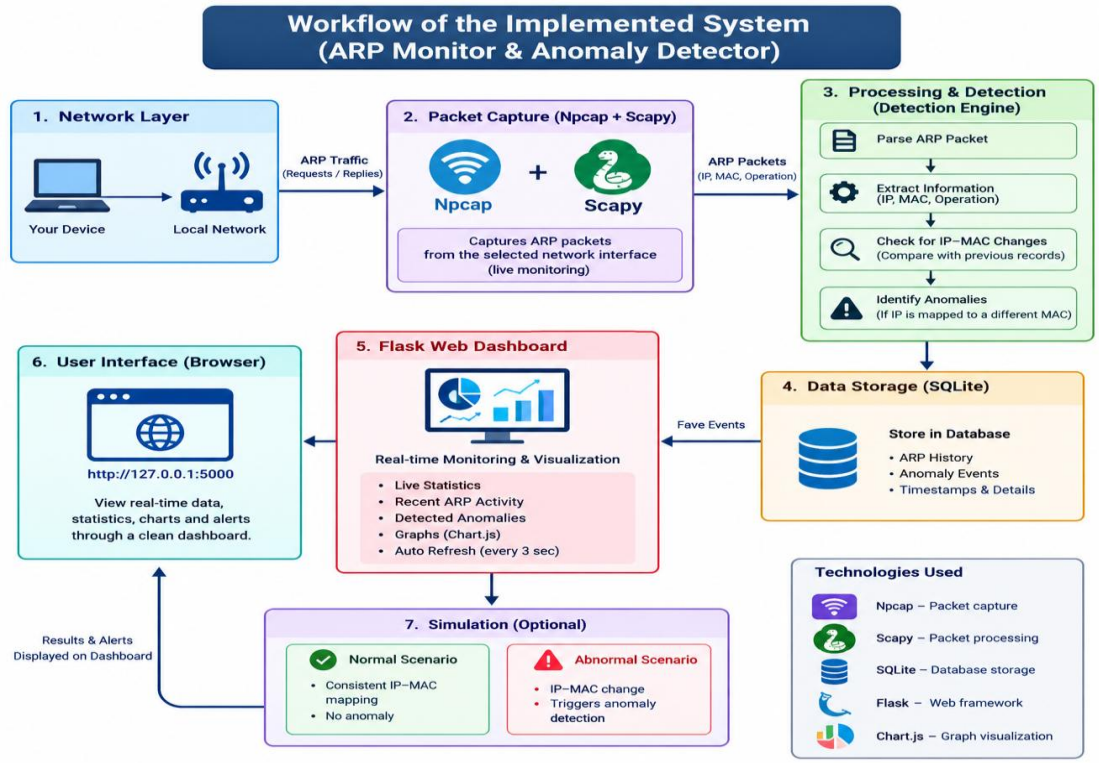
6. Existing System vs Proposed System

The table below contrasts traditional network monitoring mechanisms with the proposed ARP Monitor & Anomaly Detector system:

Existing System	Proposed / New System
Observes ARP requests and replies	Continuously monitors ARP traffic
IP–MAC information is inspected manually	IP–MAC mappings are automatically recorded
Limited historical tracking	Maintains ARP history using SQLite
IP–MAC changes require manual comparison	Automatically detects IP–MAC changes
No dedicated anomaly database	Stores detected anomalies with details

Limited statistical analysis	Provides ARP and anomaly statistics
Visualization requires separate tools	Provides an integrated dashboard with charts
Manual monitoring is common	Dashboard updates automatically
No built-in simulation	Supports normal and abnormal simulations
Results are viewed using separate utilities	Centralized Flask-based monitoring interface

7. System / Network Architecture:



ARP Monitor and Anomaly Detector is an architecture composed by several interlinked levels which are responsible for capturing, processing, storing, analyzing and visualizing ARP data. ARP starts at an individual device level within its own LAN before reaching out for presentation by means of this system via a web interface.

Architecture Flow:

Network Layer:

Network Layer is a Local Area Network (LAN), which allows several computers to exchange information through their respective IP address and MAC address. When a device needs to find out its own MAC address from an IP address it sends ARP request and reply packets. This proposal is about monitoring those ARPs over a certain type of network port.

Packet Capture Layer:

Packet Capture Layer obtains ARP packet from an active network interface. Scapy captures, decodes and processes ARP packets whereas NPCAP has a way of capturing them from within an operating system like windows. Captured packet contains details like a source IP address, source MAC address, destination info and ARP operation type.

Packet Processing Layer:

Once it has captured an ARP packet from a device on its network card, this software reads that data to extract useful details needed by our program for tracking purposes. IP Address; MAC Address; ARP Request/Reply Operation; Timestamp. The processed data will be sent by this system towards a detector where it can compare against existing datasets.

Mapping of IP to MAC address detection layer.

A detection layer keeps track of past IP-MAC pairs encountered by it. Upon receiving an ARP packet, it matches that packet to its previous MAC address in history database. A single IP address may be seen from another MAC addresses which would indicate an abnormality by this software. Then it will be saved along with its old MAC address and any other information that may be needed for reference purposes.

Data Storage Layer:

SQLite is a file-based relational database management system that stores data persistently to monitor it. ARP event storage, MAC address mapping of an IP, timestamp data for anomalies along with its description is stored here. Historical data maintenance enables comparison with past mapping records and tracking of network activities for analysis purposes.

Statistics and Analysis Layer

Data analysis helps us produce useful monitoring statistics from it. Value is calculated by this program like.

Total

Unique IP Addresses

Unique MAC Addresses

Total Detected Anomalies

Normal

Anomalous

The data is an overview to monitor network activities.

Web Application Layer

Flask is a tool that connects to an existing database of monitored information for display on this website. ARP event retrieval maps data statistics anomalies are stored within an SQLite database which is then displayed on a dash board. It is also capable to update automatically for display on screen.

Visualization Layer:

The dashboard uses Chart.js to present the collected data graphically. The ARP activity and anomaly statistics are shown as graphs so that users may better comprehend their networks' activities and detect any anomalies from them. Also tables can be used for displaying detailed ARP and anomaly data.

User Interface Layer:

A person uses an internet browser to interact with this type of monitor device. Local access to a Flask app is available at <http://localhost:5000>.

Http://127.0.0.1:5000.

A dashboard is an overview for monitoring ARP network activity, present data on it now; past history also available to detect any abnormalities among them.

8. Network Design and Implementation

The ARP Monitor & Anomaly Detector runs on a Local Area Network (LAN). It captures ARP traffic, tracks IP–MAC mappings, checks for changes, and presents the collected information through the Flask dashboard.

8.1 Network Type

The project uses a LAN with IPv4 and MAC addressing. During testing, the selected Wi-Fi interface had IP 172.23.245.26 and MAC 1c:c1:0c:ce:3a:29.

8.2 ARP Protocol

The monitor records ARP requests and replies, IP and MAC addresses, the operation type, and changes in IP–MAC mappings.

- ARP requests and replies
- IP and MAC addresses
- ARP operation type
- IP–MAC mapping changes

These ARP observations are used for monitoring and change detection.

8.3 Packet Capture

Npcap and Scapy are used to capture and process live ARP packets. The required IP, MAC, and operation information is extracted from the packets.

The monitor records ARP requests and replies, IP and MAC addresses, the operation type, and changes in IP–MAC mappings.

- ARP requests and replies
- IP and MAC addresses
- ARP operation type
- IP–MAC mapping changes

These ARP observations are used for monitoring and change detection.

8.4 IP–MAC Change Detection

Each new IP–MAC mapping is checked against the previously stored value for that IP address.

New ARP Packet

↓

Extract IP + MAC

↓

Compare with Previous Mapping

↓

Same → Normal Event

Different → Potential Anomaly

When the same IP address is found with a different MAC address, the change is recorded as a potential anomaly.

8.5 Database Storage

SQLite stores ARP events, IP–MAC mappings, timestamps, anomaly records, severity, and descriptions. The stored records provide history for later comparison.

8.6 Client–Server Communication

The Flask application handles local communication between the browser, database, and monitoring interface:

Web Browser

↓

Flask Server

↓

SQLite Database

↓

Monitoring Data

↓

Web Dashboard

The dashboard is available locally at http://127.0.0.1:5000.**http://127.0.0.1:5000**

8.7 Statistics and Visualization

The dashboard displays important statistics such as:

- Total ARP Events
- Unique IPs
- Unique MACs
- Detected Anomalies

Chart.js displays ARP activity and normal/anomalous events. The dashboard refreshes the information shown to the user.

8.8 Simulation

Two controlled cases are used:

- **Normal: The IP–MAC mapping remains consistent, so no anomaly is recorded.**
- **Abnormal: The MAC address for an existing IP changes, so a potential anomaly is generated.**

The implementation brings packet capture, IP–MAC tracking, change detection, storage, statistics, and visualization into one system.

9. Results / Expected Results

1. The ARP monitor and anomaly detector results are shown hereby.
2. Live ARP packet capture was done by npcap and scapy.
3. ARP is connected to a computer via an available network interface slot on that machine.

4. IP address extraction was done by this device for packet capture while also extracting MAC address information and type of an ARP operation on it.
5. This is an ARP event detector that detects all these events at the local level.
6. We track MAC address information of an IP address for tracking purpose and historical comparison.
7. This paper compares newly discovered IP MAC mappings with those already known.
8. We detect anomalies in changes of current IP MAC connections.
9. Old and new MAC addresses are found for anomaly detection purposes.
10. SQLite databases are used to store information about an ARP event like its location map coordinates time stamps anomaly reports etc.
11. A Flask-based web dashboard is a useful way of monitoring network activity at one location.
12. Dashboard showed total ARP events, unique IP addresses, unique MAC address count and anomalies.
13. The ARP activity and monitoring information are shown as a chart by Chart.
14. The automatic updating function of a dashboard is able to provide current information about it through an interface or application software program.
15. Normal and abnormal scenarios were simulated to demonstrate a constant mapping process along with an IP MAC change.

For instance, you can capture an ARP packet for analysis through MAC address matching that helps identify database anomalies.

Overall Result

It worked well on an active Wi-Fi network where it could detect live ARP queries from other devices. This paper shows how integrating real-time ARP surveillance systems with past data on MAC addresses, automatic changes detection software to store information about them databases analysis of statistics online interface helps monitor traffic patterns within an organization's LANs effectively detect any unusual behavior between those two components such as mismatches between known devices' names and their actual hardware address assignments or unauthorized access attempts by hackers etc.

10. Conclusion

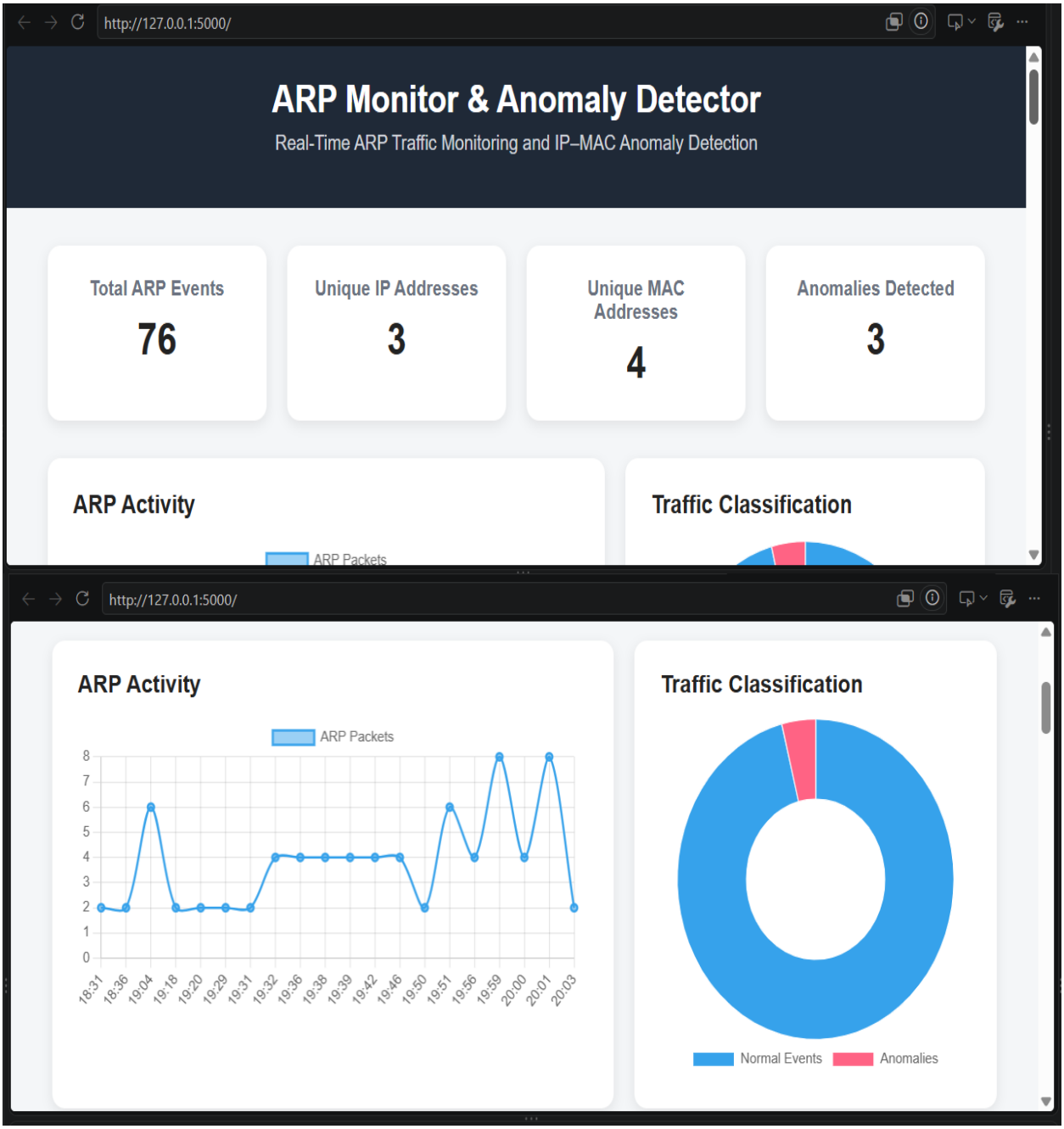
ARP Monitor and Anomaly Detector is an effective solution to monitor ARP traffic on your LAN. ARP is important for communication as it maps IPv4 addresses to their corresponding MAC addresses; however, changes in these mappings can provide useful indications of unusual network behavior. This proposal is based on an ARP behavior model that will be used to continuously monitor it and analyze its performance.

ARP request/response is captured by this device through npcap and scapy where it retrieves details like IP address, MAC address, and type of ARP operation. IP-mac addresses of this system can be stored as a table within an sqlite file for comparison purposes. If there are any changes to an existing IP address, they will be recorded by our system as anomalies along with information like its old MAC address, current MAC address, time of detection and description.

Flask based web application is an integrated tool to view ARP traffic data and monitor it. Displays all ARPs, each IP address is counted once while a MAC address will be shown separately from its corresponding anomaly chart graphically via Chart. It is used for simulating both normal and abnormal situations so that we can test out our detection algorithm under different conditions.

In general, this paper is an extension to the basic idea for resolving ARPs through constant tracking, MAC address histories, automatic changes detection mechanism, anomaly identification technique, data base management system implementation statistics analysis tool and a web based graphical display function. This shows us that basic networking principles are combined with technology for creating an effective monitoring system of networks. It is also an ideal base to develop new features like better anomaly detection algorithms, enhanced warning systems and deeper insights on networks.

11. OUTPUT


http://127.0.0.1:5000/
ARP Monitor & Anomaly Detector
Real-Time ARP Traffic Monitoring and IP–MAC Anomaly Detection
Total ARP Events
76
Unique IP Addresses
3
Unique MAC Addresses
4
Anomalies Detected
3
ARP Activity
ARP Packets
Traffic Classification
http://127.0.0.1:5000/
ARP Activity
ARP Packets
8
7
6
5
4
3
2
1
0
18:31
18:36
19:04
19:18
19:20
19:29
19:31
19:32
19:36
19:38
19:39
19:42
19:46
19:50
19:51
19:56
19:59
20:00
20:01
20:03
Traffic Classification
Normal Events
Anomalies

Recent ARP Activity

Time	IP Address	MAC Address	Operation	Interface
2026-10-02 20:03:38	172.23.245.26	1c:c1:0c:ce:3a:29	reply	live
2026-10-02 20:03:38	172.23.245.85	de:81:87:5f:61:e6	request	live
2026-10-02 20:01:30	172.23.245.26	1c:c1:0c:ce:3a:29	reply	live
2026-10-02 20:01:30	172.23.245.85	de:81:87:5f:61:e6	request	live
2026-10-02 20:01:30	172.23.245.26	1c:c1:0c:ce:3a:29	reply	live
2026-10-02 20:01:30	172.23.245.85	de:81:87:5f:61:e6	request	live

Detected Anomalies

Time	IP Address	Old MAC	New MAC	Severity	Description
2026-10-02 18:36:48	192.168.1.50	AA:BB:CC:11:22:33	11:22:33:44:55:66	**HIGH**	IP 192.168.1.50 changed from AA:BB:CC:11:22:33 to 11:22:33:44:55:66
2026-10-02 18:36:48	192.168.1.50	11:22:33:44:55:66	AA:BB:CC:11:22:33	**HIGH**	IP 192.168.1.50 changed from 11:22:33:44:55:66 to AA:BB:CC:11:22:33
2026-10-02 18:31:38	192.168.1.50	AA:BB:CC:11:22:33	11:22:33:44:55:66	**HIGH**	IP 192.168.1.50 changed from AA:BB:CC:11:22:33 to 11:22:33:44:55:66

12.GIT REPOSITORY:

https://github.com/kusuma-kr/ARP-Monitor-Anomaly-Detector.git

